

STOP HUNTING DATA IN PDFS

A working parser, built in Excel. What it took, and where it broke.



Copy. Paste. Check. Repeat 340 times.

Input and output tax invoices arrive as government PDFs. Every line has to land in a sheet an accountant can reconcile. Doing that by hand is not difficult work — it is just relentless.

WHAT IT ACTUALLY COST

340

invoices in a
monthly batch

6 hrs

of transcription
per batch

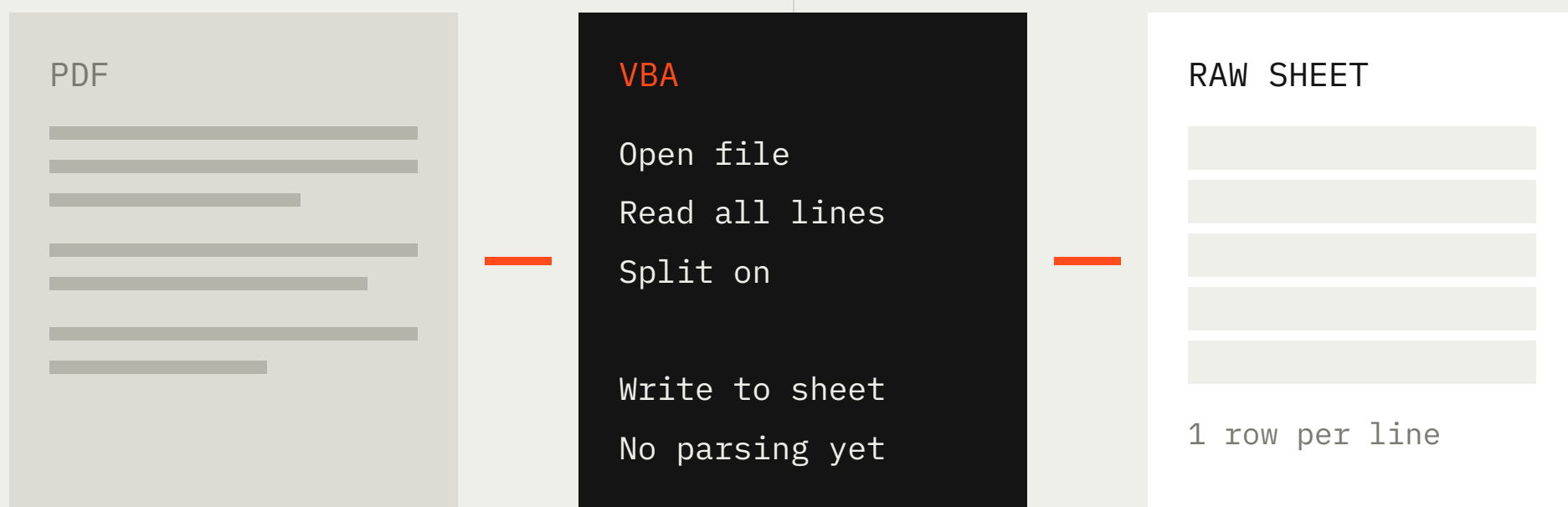
~20%

keying errors
found later

The 2% was the real problem. Nobody knew which rows were wrong.

Extract everything. Understand it later.

The first version doesn't try to be clever. It dumps the whole PDF into a raw sheet, line by line, exactly as it comes.



WHY THIS ORDER

Separating extraction from interpretation means a layout change breaks one step, not the whole macro.

Don't trust position. Trust the label.

Row 14 is only row 14 until an invoice has one extra line. Find the word, then take what sits beside it.

CELL	RAW VALUE	
A11	FAKTUR PAJAK	
A12	Nomor Seri	
A13	Dasar Pengenaan Pajak	← anchor
B13	12.500.000	← the value
A14	PPN	

THE TRADE-OFF

Anchors survive layout drift. They do not survive a vendor renaming the field — and one eventually will.

Write the rule once. Point it at anything.

Module1.bas

```
Function ParseField(sheet, anchor)
    row = FindRow(sheet, anchor)
    If row = 0 Then Exit Function
    ParseField = Clean(sheet(row, 2))
End Function
```

CALLED LIKE THIS

ParseField(s, "Dasar Pengenaan Pajak")	→	12500000
ParseField(s, "PPN")	→	1250000
ParseField(s, "Nomor Seri")	→	010.003-24

Three fields, one function. Adding a fourth is a line of config, not a new macro.

Automation is mostly anticipating human inconsistency.

The parsing was the easy half. These four things caused every failure worth naming.

Scanned, not text

A photocopy has no text layer at all. The macro reads nothing and reports nothing — the worst failure mode.

The extra line

One vendor adds a second address row. Every fixed row reference below it now points at the wrong cell.

1.250,00 or 1,250.00

Decimal separators flip between sources. Parse it wrong and you are out by a factor of a hundred.

Silent empties

A missing field returns blank, not an error. Blank looks like zero on a reconciliation sheet.

Rule: fail loudly. A parser that returns nothing must say so.

A clean table, ready to reconcile.

NOMOR SERI	DPP	PPN	STATUS
010.003-24.1102	12,500,000	1,250,000	OK
010.003-24.1103	8,400,000	840,000	OK
010.003-24.1104	—	—	SCANNED
010.003-24.1105	22,000,000	2,200,000	OK

The flagged row is the point. It isn't a failure — it's the one invoice a human still needs to open.

AFTER

6 hrs

became

9 min

per 340-file batch

7

flagged for a human

The parser was easy. The exceptions were the project.

Anyone can extract a clean PDF. What you are really building is a system that knows when it is wrong — and says so before the numbers reach a spreadsheet.

Extract first, interpret second

Anchor on labels, never on row numbers

A blank is not a zero — flag it

Which repetitive document task could you parse next?